GHraisoni COLLEGE
Engineering and Management
Nagpur

RSOC-2026

RSOC 2026

GROUP 12

- Domain :– AI/ML
- Problem Statement Title:- Bug Detector Intelligent Code Analysis Tool
- Team Leader Name: - Arya Pathrabe
- Group No:- 12
- Team Name:- The Forgers
- Members:- Arya Pathrabe
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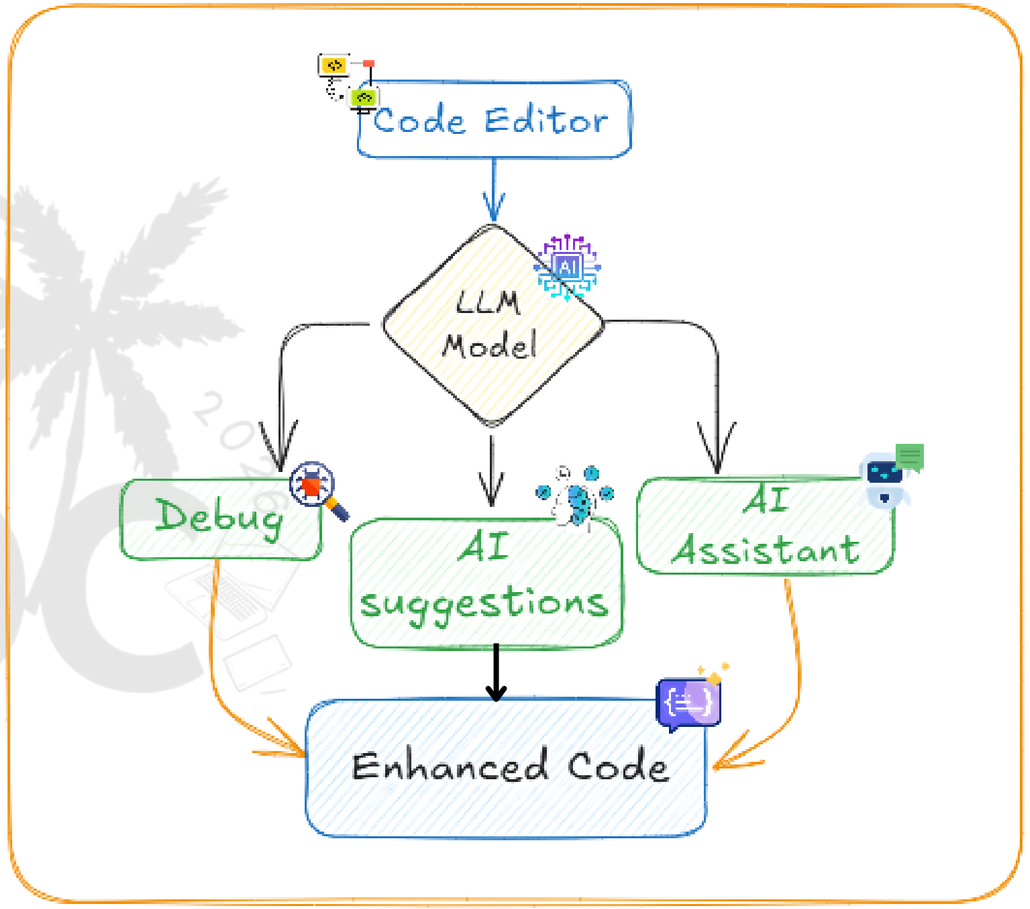
IDEA TITLE

Proposed Solution: **CodeStrix** AI-Powered Code Analysis Platform

- Web-based editor supporting **WebDEV, Python, JavaScript, Java, C++**
- Detects logical bugs, inefficient algorithms, runtime risks
- Provides AI explanations and **optimized code fixes**
- Highlights issues **directly in the code editor**
- Developer dashboard with **analytics & history tracking**

TECHNICAL APPROACH

- **Web-based AI Code** Analysis Platform
- Code input through **Monaco Editor**
- Static Code Analysis using tools like **ESLint**
- AI Model Integration **(Gemini / Code LLM)** for detecting logical bugs and inefficiencies
- Results processed and displayed with severity levels and confidence scores
- Developer **dashboard visualizes** code quality metrics and issue reports



PROTOTYPE :

FEASIBILITY AND VIABILITY

Technical Feasiability

- Built using modern **web technologies (React, Node/Python**) and **scalable AI APIs**.
- Compatible with **low-end devices** and **offline conditions.**

Market Viability

- Growing demand for **AI-assisted developer tools.**
- Useful for software **developers, students,** and **development teams**.

Implementation Feasibility

- Web-based platform accessible **through any browser.**
- Requires **NO setup and easy integration** with developer workflows.

IMPACT AND BENEFITS

Social Benefits

- Helps learners improve coding skills
- Encourages writing clean and efficient code

Economic Benefits

- Reduces development time and debugging costs
- Improves efficiency in software development projects

Competitive Advantage

- Combines AI reasoning + static code analysis.
- Provides confidence scores, issue severity, and improvement suggestions.

RESEARCH AND REFERENCES

Tools & Technologies Referenced

- **Monaco Editor** – Web-based code editor with syntax highlighting
- **ESLint –** Detects common JavaScript code issues
- **Ruff** – Fast Python code quality checker
- **Clang Static Analyzer** – Detects bugs in C/C++ programs
- **Gemini – AI** reasoning for code analysis